Analysis report

1) In the original scenario, the satisficing plan contains 31 actions, whereas the optimal plan contains 25 actions. In the extended scenario, the satisficing plan contains 60 actions, whereas the optimal plan contains 40 actions.

The optimal planner produces shorter plans because it minimizes the total number of actions, while the satisficing planner focuses on finding a valid solution without guaranteeing the minimum plan length. The extended scenario requires more actions because two additional rocks are introduced and the direct connection between Base and SiteB is closed due to the dust storm.

2) The rover has two storage bays, Bay1 and Bay2, so it can carry a maximum of two rocks at a time. It collects two rocks in separate bays and returns to Base to deposit them before collecting more rocks. This reduces unnecessary trips to Base.

The rover picks up the drill at SiteE before collecting any hard rocks. The drill is required to collect Rock3, Rock5 and Rock7. After collecting these rocks, the rover deposits them at Base using the two storage bays.

3) In the extended scenario, the dust storm closes the direct connection between Base and SiteB in both directions. Therefore, the rover reaches SiteB through an alternate route: Base -> SiteA -> SiteC -> SiteD -> SiteB.

The rover uses the one-way connection from SiteC to SiteD. It cannot travel directly from SiteD to SiteC, but it can return through SiteD → SiteE → SiteC → SiteA → Base. Thus, the rover reaches both SiteB and SiteD while following the available connections.